

Project Specification for a Messaging Application

Group Technical Thinking and Writing – Group 1.8

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Section A: Problem Definition

A1. Problem statement & Project aims

Aims:

The project aims to create a messaging application that enables people to chat in real time. The application should provide a user-friendly interface so that the users have a good experience of using it. The application should be secure and private in terms of storing users' data and chat history. It should offer a variety of additional features such as video calls and voice messages that allow individuals and groups to express themselves according to their preference. It should support high volumes of users simultaneously avoiding system crashes.

Objectives

A1.1 - To send and receive a message within milliseconds.

A1.2 - To use an encrypted authentication system to protect accounts from unauthorised access.

A1.3 - To develop an end-to-end encryption that keeps chat history private to sender and receiver.

A1.4 - To develop features: text message, group chat, images and voice message.

A1.5 - To develop high-quality video calls.

A1.6 - To ensure the application can be accessible on IOS, Android, and PCs.

A1.7 - To design a user interface that provides customization options such as notification settings.

A1.8 - To use a server that can manage communication between 10000 users simultaneously.

A2. Feasibility Study and issues on Ethics and Sustainability

A2.1 - The service should be able to function 24/7 duration.

A2.2 - Long-term maintenance plan such as regular updates for security patches and features.

A2.3 - Technical: Compatibility across platforms – solution: Cloud computing platforms

A2.4 - The energy consumption for loading the app should be low. This will be achieved by using renewable power sources.

A2.5 - The application should gain consent from the user to use their data for improving user experience, in compliance with GDPR regulations.

Feasibility Study

Marketing Segment:

The messaging application market is highly competitive yet rapidly expanding. According to a report by Statista in 2023, the global Number of mobile phone messaging app users worldwide from 2022 to 2030 is predicted to grow by 192% (Statista, 2023). Currently, popular applications like WhatsApp and Telegram dominate the market by offering a blend of secure, user-friendly features and cross-platform functionality.

The proposed messaging application differentiates itself by combining innovative user experiences, such as face filters and advanced privacy controls, with robust scalability to handle large user volumes and strong encryption facilities. Targeting niche markets, such as business users who value security or younger demographics seeking creative engagement features, provides opportunities for market penetration. Additionally, compliance with GDPR and similar regulations allows it to operate in privacy-conscious regions like the European Union.

The demand for real-time communication tools, particularly post-pandemic, suggesting a significant opportunity to capture users looking for secure, multi-platform messaging solutions. Therefore, the market demand is very large, so our project is feasible and can fully meet these demands.

Technical segment:

The technical feasibility of the project lies in integrating cutting-edge encryption protocols and scalable backend infrastructure. End-to-end encryption ensures the privacy of conversations, safeguarding against cyber threats. This is achievable using existing frameworks such as TLS or proprietary implementations of distributed proxy re-encryption protocols (Demir & Altilar 2024).

A scalable database architecture which is optimised for high concurrency, is essential to handle simultaneous communications for thousands of users. In addition, as global userbase grows a global network of servers will provide consistent speeds for users around the world.

Cross-platform compatibility using frameworks like React Native or Flutter ensures a consistent user experience across Android, iOS, and desktop platforms (Masoodi & Andrey, 2022). Video calling functionality, employing low-latency WebRTC protocols, enhances the application's appeal by offering real-time, high-quality communication.

Our project is very feasible in terms of market readiness, technological availability. However, it also requires a large amount of upfront capital investment. This can be done by negotiating with venture capitalists in combination with a sequence of pre-seed and seed round funding being done while initial development is happening. This will allow a streamlining of development and securing the capital needed to deploy the infrastructure necessary to deliver our messaging app to users around the world. To secure funding in this manor, we will demonstrate our income streams to investors. Our main stream of income will come from in-app advisements. However, additional income streams will be generated by in-app purchases which will allow users to subscribe monthly to gain access to exclusive features such as the removal of some adverts inside the app and increased personalisation.

A3. User Requirements

A3.1 - Communicate with friends and family remotely at any time, from anywhere in the world.

A3.2 - Convenient interface and various features to increase interaction such as face filters.

A3.3 - The application should be secure to keep data private.

A3.4 - The application should allow group chat functionality.

A3.5 - The application can be used on all devices and different systems such as IOS, Android and desktop. (Rawat et al., 2024)

A4. Constraints

A4.1 - The user experience will be restricted by the number of available platforms they can access to a messaging application. For example, Android and IOS. Minimum requirements for device.

A4.2 - The backup storage space requirement must be compatible to avoid the loss of message records.

A4.3 - The core communication function should be focused.

A4.4 - The power system should work sufficiently to avoid cache. E.g. Many users at the same time.

Section B: System Requirements

B1. Input / Output Requirements

B1.1 - End to end encryption

B1.2 - Inputs to the application are provided by keyboard for users to send texts and emojis.

B1.3 - Inputs include voice message and video calls are provided by microphone, sensors and camera.

B1.4 - Inputs to the network are provided by WIFI and data package.

B1.5 - Account registration requires keyboard to input username and password, and log in process requires biometric data such as face recognition.

B1.6 - Access device media to send photos, videos and documents.

B1.7 - Incoming messages or calls are processed by the server.

B1.8 - Notification is received via sound and banner display.

B2. Operational Requirements

B2.1 - Login authentication: multi-factor mechanisms are applied to verify user identity while they are logging in. End-to-end authentication guarantees security of user data.

B2.2 - Real-time message: users can send and receive messages instantly within milliseconds. The message is temporarily stored if recipients are offline.

B2.3 - Sharing: the user can access their device media to select files, photos, videos that they want to share.

B2.4 - Video calls: the application should minimise latency and provide high-quality video calls.

B2.5 - Privacy control: the users should be able to manage their privacy settings including profile picture and states. They should be able to block and unblock other users.

B2.6 - Application update: users are notified of a new version of the application and option to update and download it. Information about what is updated and how long it will take should be provided.

B2.7 - Using the User-Centred Design (UCD) method which considers the feedback from the users in every design process. Getting data directly from users enables immediate and accurate update of the features and system. (Zahida et al., 2021)

B3. Functional Requirements

B3.1 - User authentication: users are requested to register account and then set two-factor authentication (2FA) that uses QR code and Fingerprint/ Face recognition to log in in the future. The users can log in quicker and do not have to memorise their password. QR code allows storing large amounts of data and fast reading. Fingerprint /face recognition is used as another authentication method. (Segoro et al., 2020)

B3.2 - Security: protect conversation and chat history by applying end-to-end encryption to ensure that only the sender and intended recipients can read the message. An additional distributed proxy Re-Encryption Protocol is used to increase security and privacy. With CKKS schemes based on RLWE construction, the main data owner generates the individual public key first and then the other data owners generate their public keys from the first data owner's public key. This means that the semi-trusted server can only obtain a public re-encryption key, and when the client sends more than one query, the re-encryption process is performed instead of distributed decryption for each query. (Demir & Altılar, 2024)

B3.3 - Messaging: Provide one-on-one and group chat. Having various features to make chat fun and realistic. High-quality video calls display face-to-face conversations. Emojis create interesting interactions between each other. Voice messages allow users with different reading and writing abilities to use the application, and it is a more convenient way of sending and listening to messages.

B3.4 - User interface: Using languages HTML, CSS, JavaScript to create chat interface, Python to focus on creative of the application. Having an attractive, innovative, and many-choice UI /UX design to enhance user experience. Theme customizations give users a choice on chat interface. (Rawat et al., 2024, pp. 1-6)

B3.5 - The sign-up process should require minimal information related to the user. The account creation process should not rely heavily on Personal Identity Information. This ensures that users are not affected as much in case of any data leaks, helping to maintain privacy.

B3.6 - Chat History: the application should be able to store chat history in the database management system. This will also allow users to download their data.

B4. Non-functional Requirements

B4.1 - The ability to handle large numbers of users at the same time. (e.g. 1 Million)

B4.2 - The message should be delivered fast, within milliseconds.

B4.3 - The application should work 24/7 duration.

B4.4 - Compliance with GDPR or similar data protection regulations.

B4.5 - End-to-end encryption for all communications.

B4.6 - Regular maintenance and updates for security and features.

Section C: Key Design Challenges

C1 - User Engagement

For data-driven businesses, personalized marketing greatly increases productivity by encouraging real customer interactions. Essential aspects that increase user satisfaction, encourage conversions, and foster brand loyalty are provided by messaging applications. Innovation is important in this situation because companies can use interactive technologies like augmented or virtual reality (AR and VR), games, chatbots, and branded stickers to engage consumers. These innovative resources not only improve user experience but also give companies distinctive ways to stand out in a crowded market.

C2 - Enhanced Communication and Customer Engagement

Messaging applications employ chatbots, push notifications, and in-app messaging to facilitate direct and personalized communication between businesses and consumers. These tools enable companies to provide immediate updates, recommendations, and support, fostering real-time interactions that enhance customer relationships and satisfaction. Proactive engagement through these channels can significantly improve communication strategies, leading to increased customer loyalty and engagement.

For instance, AI-powered chatbots are increasingly utilized in customer service to deliver instant assistance, thereby improving user satisfaction and engagement. These chatbots can handle a variety of tasks, from answering frequently asked questions to guiding customers through complex processes, effectively enhancing the overall customer experience (Gupta and Khan, 2024).

Moreover, the integration of push notifications and in-app messaging allows businesses to proactively reach out to customers with relevant information and personalized offers. This approach not only keeps customers informed but also fosters a sense of connection and engagement with the brand (McKinsey & Company, 2023).

C3 - Annoyance and Disturbance

Spam messages cause significant annoyance and disturbance to users. They disrupt daily routines, creating an unpleasant user experience. This persistent interference frustrates users and detracts from their ability to focus on important messages. (Thirumalai et al., 2024)

C4 - Financial Scams and Security Threats

People are tricked by some spam messages into falling for financial scams or fraudulent schemes, which results in losses. The dissemination of malware and phishing attempts can also be facilitated by spam mails. Link clicking and replying to such communications can lead to security lapses and jeopardize the integrity of users' data and devices.

(Thirumalai et al., 2024)

C5 - Verification

Verification via email the account activation link should be given to the user's (supplied) email address, and it should be completed as part of the account creation procedure.

(Akram and Ko, 2024)

C6 – Scalability problem

The database architecture used in the current instant messaging & presence architecture is known as the master-slave architecture

Increasing the number of slaves does not decrease the load of replicated update queries per slave, as each slave needs to process all updates. With an increasing write load, the resources left at the slave nodes to serve read requests become smaller and smaller, thus increasing the response time of the system (Schippers, J. 2009).

C7 – Online disinformation

Online disinformation, as an area of inquiry, is still very much in its infancy. More recently, there has been a move by some to develop more critical approaches to understand disinformation in efforts to upend preexisting assumptions, including that a healthy information ecosystem and clear lines between “true” and “false” had existed before the advent of social media (M. J. Masoodi and S. Andrey, 2022).

C8 – Cross platform compatibility

The messaging app should provide a seamless and consistent user experience across various devices and operating systems which can be done by utilising frameworks like React Native, Flutter, or Xamarin to write a single codebase that can deploy on multiple platforms. The app should also be optimised for various screen sizes and resolutions.

References

- Akram, N. and Ko, R. (2014). *End-to-End Secure and Privacy Preserving Mobile Chat Application*. [online] Available at: <https://pure.royalholloway.ac.uk/ws/files/23295486/Paper.pdf> [Accessed 6 Dec. 2024].
- Demir, B.B. and Altılar, D.T. (2024). Distributed Proxy Re-Encryption Protocol for Secure Multiparty Computation with Fully Homomorphic Encryption. *2024 IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)*. [online] doi: <https://doi.org/10.1109/blackseacom61746.2024.10646222>
- Dierks, T. and Rescorla, E. (2008). The Transport Layer Security (TLS) Protocol Version 1.2. doi: <https://doi.org/10.17487/rfc5246>
- Gupta, Y. and Farheen Mujeeb Khan (2024). Role of artificial intelligence in customer engagement: a systematic review and future research directions. *Journal of modelling in management*. doi: <https://doi.org/10.1108/jm2-01-2023-0016>
- Masoodi, M.J. and Andrey, S. (2022). Understanding the Use of Private Messaging Apps in Canada and Links to Disinformation. *IEEE Technology and Society Magazine*, 41(3). doi: <https://doi.org/10.1109/mts.2022.3197115>
- McKinsey (2023). *AI Customer Service for Higher Customer Engagement | McKinsey*. [online] www.mckinsey.com. Available at: <https://www.mckinsey.com/capabilities/operations/our-insights/the-next-frontier-of-customer-engagement-ai-enabled-customer-service> [Accessed 7 Dec. 2024].
- Rawat, V., Jain, P., Tomar, A., Piyush Guzar, Negi, V. and Gupta, V.K. (2024). A Web-Based Real Time Chat Application for User Friendly Interaction. [online] doi: <https://doi.org/10.1109/acet61898.2024.10730503>
- Schippers, J., Remke, A., Punt, H., Wegdam, M. and Haverkort, B. (2010). A Massively Scalable Architecture For Instant Messaging & Presence. *Electronic Notes in Theoretical Computer Science*. doi: <https://doi.org/10.1016/j.entcs.2010.01.008>
- Segoro, M.B. and Wibowo Putro, P.A. (2020). Implementation of Two Factor Authentication (2FA) and Hybrid Encryption to Reduce the Impact of Account Theft on Android-Based Instant Messaging (IM) Applications. *2020 International Workshop on Big Data and Information Security (IW BIS)*. doi: <https://doi.org/10.1109/iwbis50925.2020.9255501>

Statista. (2023). *Mobile chat app users 2030* | Statista. [online] Available at: <https://www.statista.com/statistics/1401734/mobile-messenger-user-worldwide/> [Accessed 8 Dec. 2024].

Thirumalai, K.G., Prakash, K.S., Abirami, A.M., Ramanujam, E. and Sumitra, S. (2024). XAI-based Feature Selection for SMS Spam Classification in Dravidian Languages. *2024 5th International Conference on Innovative Trends in Information Technology (ICITIIT)*. [online] doi: <https://doi.org/10.1109/icitit61487.2024.10580065>

Zahida, W., Effendy, V. and Hadikusuma, A. (2021). *Improving Messenger Accessibility for Elderly Users using User Centered Design (UCD) Methods (Study Case: WhatsApp)*. [online] IEEE Xplore. doi: <https://doi.org/10.1109/ICSECS52883.2021.00056>